SilaiPro

Tailor & Boutique Management System

Product Requirements Document (PRD)

Version	v1.0 – Initial Release
Date	2025
Platform	Flutter (Mobile + Tablet) + Java Spring Boot
Target Users	Tailors, Boutique Owners, Staff
Status	Draft – For Review

1. Executive Summary

SilaiPro ek complete digital management solution hai jo tailors, boutique owners, aur unke staff ke liye design kiya gaya hai. Ye system customer measurements ko digitally store karta hai, orders ko track karta hai, billing aur accounts manage karta hai, aur staff ko role-based access deta hai.

Ye app Flutter framework mein build hogi jo Android phones (portrait) aur tablets (landscape) dono pe achhi tarah se kaam kare. Backend Java Spring Boot REST APIs use karega jo secure, scalable, aur maintainable ho.

Key Metric	Detail
App Name	SilaiPro
Frontend	Flutter (Dart) – Android & iOS
Backend	Java Spring Boot + MySQL
Target Devices	Phone (Portrait) + Tablet (Landscape)
Access Control	Role-Based (Admin, Manager, Staff, View-Only)
Offline Support	Partial (local cache for measurements)
Print Support	PDF generation + Bluetooth Thermal Printer
Messaging	SMS / WhatsApp integration
Payments	Manual entry (no payment gateway)

2. Problem Statement & Goals

2.1 Current Challenges

Aaj bhi bahut se tailors aur boutique owners apne records paper mein ya simple registers mein rakhte hain. Isse kai problems hoti hain:

- Customer ke naap (measurements) dhundhne mein time lagta hai ya khoj jaate hain
- Bills manually likhne padte hain, galtiyan hoti hain
- Partial payment ka koi systematic record nahi hota
- Staff ko pata nahi chalta ke kaunsa order kab deliver hona hai
- Customer ki purani history easily available nahi hoti
- Business ke accounts ka clear picture nahi milta

2.2 Solution Goals

- Ek centralized digital system banana jo sab kuch ek jagah rakh sake
- Role-based access se security ensure karna
- Customer measurements dynamic aur configurable rakhna
- Billing aur partial payment tracking automate karna
- Print aur messaging se customer experience improve karna
- Staff productivity aur accountability badhana

3. Role-Based Access Control (RBAC)

System mein 4 roles honge. Har role ke alag permissions hain jo admin backend se manage kar sakta hai.

Feature / Module	Admin	Manager	Staff	View Only
Dashboard	✅ Full	✅ Full	✅ Limited	✅ Read Only
Customer Add/Edit	✅	✅	✅	❌
Customer Delete	✅	✅	❌	❌
Measurements Add/Edit	✅	✅	✅	❌
Measurements View/Print	✅	✅	✅	✅
Orders Create/Edit	✅	✅	✅	❌
Orders Delete	✅	✅	❌	❌
Billing Create	✅	✅	✅	❌
Payment Record	✅	✅	✅	❌
Accounts View	✅	✅	❌	❌
Accounts Reports	✅	✅	❌	❌
Send Messages	✅	✅	✅	❌
Staff Management	✅	❌	❌	❌
System Settings	✅	❌	❌	❌
Measurement Fields Config	✅	✅	❌	❌

3.1 Role Management

- Admin sirf nayi roles create kar sakta hai ya existing ka naam aur permissions change kar sakta hai
- Ek user ka role Admin UI se change ho sakta hai
- Staff ka login phone number + PIN ya email + password se hoga
- Session JWT tokens se manage hoga – automatically expire after configurable time

4. Customer Module

4.1 Customer Profile

Har customer ka ek complete digital profile hoga jisme following information store hogi:

Field	Description
Full Name	Customer ka poora naam (required)
Phone Number	Primary contact – SMS ke liye use hoga (required)
WhatsApp Number	Optional – alag number ho to
Email	Optional
Address	Full address with city/area
Profile Photo	Optional camera/gallery se
Date of Birth	For special occasion reminders
Customer Type	Regular / VIP / Wholesale (configurable)
Reference	Kaun le ke aaya (referral source)
Notes / Remarks	Special instructions ya preferences
Date Joined	Auto-filled
Customer ID	Auto-generated unique ID

4.2 Customer History

Har customer ki profile mein ek comprehensive history tab hogi:

- Saari orders ka list with status, date, amount
- Total business done (lifetime value)
- Pending payment amount
- Measurement history – kaun sa naap kab set hua
- Messages sent history
- Payment transactions history

4.3 Customer Search & Filter

- Name, phone number, customer ID se search
- Filter by: customer type, area, pending payment, date range
- Sort by: name, last visit, pending amount, total orders
- Export customer list to PDF/Excel (Admin only)

5. Measurements Module (Dynamic Naap System)

5.1 Dynamic Fields – Backend Se Configure

Ye system ka ek important feature hai. Measurement fields hard-coded nahi hain – Admin backend se apni zaroorat ke hisaab se fields add, edit, ya delete kar sakta hai.

- Admin login karke 'Measurement Templates' section mein jaata hai
- Wahan naap ke categories bana sakta hai – jaise 'Gents Suit', 'Ladies Salwar', 'Sherwani', 'Kids', etc.
- Har category ke andar custom fields add kar sakta hai – jaise 'Length', 'Chest', 'Waist', 'Hip', 'Sleeve', 'Collar', etc.
- Field type choose kar sakta hai: Number (cm/inch), Text, Select (dropdown), Notes
- Field order drag-drop se change kar sakta hai
- Optional ya required mark kar sakta hai
- Unit preference set kar sakta hai – CM ya Inch (globally ya per customer)

5.2 Measurement Entry Screen

Feature	Description	Platform
Template Selection	Customer ke liye konsa garment type hai choose karo	Phone + Tablet
Quick Entry	Side-by-side list – field naam aur number input	Tablet (Landscape)
Single Column	Scroll karte hue fields – ek ek karo	Phone (Portrait)
Body Diagram	Visual body outline pe tap karke fields navigate karo	Optional Feature
Voice Input	Speak karo aur naap fill ho – flutter speech plugin	Phone
Comparison View	Aaj ka naap vs pichle naap ka diff dikhao	Both
History Timeline	Kab kab naap liya – timeline view	Both

5.3 Print – Naap Ka Print

Staff ya Admin kisi bhi customer ke measurements print kar sakta hai:

- PDF generate hogi measurements ki – professional layout
- Customer ka naam, ID, phone, garment type, measurements sab include honge
- Shop ka naam, logo, contact details bhi header mein hoga
- Print directly to Bluetooth thermal printer ya share as PDF
- Multiple garment types ek hi print mein aa sakte hain
- Date aur staff ka naam bhi print hoga (accountability ke liye)

6. Orders Module

6.1 Order Creation

Field	Description
Order ID	Auto-generated (e.g., ORD-2025-001)
Customer	Existing customer select ya naya add
Garment Items	Multiple items add kar sako – ek order mein kai kapde
Description	Har item ke liye details – fabric, color, design
Measurements	Customer ke saved measurements link ya new measurements
Quantity	Kitne pieces
Price Per Piece	Manual entry
Total Amount	Auto-calculated
Advance Payment	Order time pe liya paise
Expected Delivery	Date picker
Priority	Normal / Urgent / Express
Special Instructions	Koi khas baat
Trial Date	Fitting/trial ki date (optional)
Assigned To	Konsa tailor ya staff handle karega
Status	Received → Cutting → Stitching → Trial → Ready → Delivered

6.2 Order Status Workflow

Har order ek defined workflow follow karta hai:

Order Status Flow
1. ORDER RECEIVED → Order create hoti hai, advance liya
2. IN CUTTING → Fabric cut ho raha hai
3. IN STITCHING → Silai chal rahi hai
4. TRIAL PENDING → Customer ko trial ke liye bulana hai
5. ALTERATION → Trial ke baad changes
6. READY → Delivery ke liye taiyar hai
7. DELIVERED → Customer ko de diya, final payment
8. CANCELLED → Order cancel

6.3 Order Dashboard

- Today's orders – kya deliver hona hai aaj
- Overdue orders – jinki delivery date nikal gayi
- Status-wise grouping – kitne orders kahan hain
- Quick status update – staff swipe ya tap se status change kare
- Order search by customer name, order ID, date, status

7. Billing & Invoice Module

7.1 Invoice Generation

Har order ke liye ek proper invoice ban sakti hai:

- Invoice number auto-generate (INV-2025-0001)
- Customer ki details automatically pull hoti hain
- All order items with quantity, rate, amount list
- Subtotal, discount, total amount
- GST/Tax option (optional – enable/disable from settings)
- Advance paid deduct karke balance due dikhayega
- Invoice date aur due date
- Payment terms / notes section

7.2 Partial Payment Tracking

Ye ek core feature hai. Agar kisi ka total bill Rs. 20,000 hai aur usne Rs. 10,000 diye hain:

Payment Record Field	Description
Invoice Total	Rs. 20,000 (total bill)
Payment 1	Rs. 10,000 – Date: 15 Jan – Mode: Cash – Note: Advance
Remaining Balance	Rs. 10,000 (auto-calculated)
Payment 2	Rs. 5,000 – Date: 20 Jan – Mode: UPI – Note: Partial
Remaining Balance	Rs. 5,000 (live updated)
Payment Status	Partial / Paid / Overdue (auto-update)
Reminder Flag	Manually mark karo – remind karna hai

Har payment record mein:

- Amount paid
- Date of payment
- Payment mode: Cash, UPI, Bank Transfer, Cheque (manual entry)
- Staff ka naam jo payment liya
- Optional receipt number
- Notes/remarks

7.3 Print Invoice

- Professional PDF invoice generate karo
- Shop logo, name, address, GST number include

- Customer ke details, order items, payment history sab
- Thermal printer (58mm/80mm) ke liye compressed layout
- A4 print ke liye full layout
- Share via WhatsApp ya email directly app se

8. Accounts & Finance Module

8.1 Dashboard Overview

Admin aur Manager ke liye financial summary:

Card	Today	This Month	This Year
Total Revenue	₹ X	₹ X	₹ X
Amount Received	₹ X	₹ X	₹ X
Pending Dues	₹ X	₹ X	₹ X
Orders Created	N	N	N
Orders Delivered	N	N	N

8.2 Accounts Reports

- Daily collection report – date-wise kitna aaya
- Pending dues report – konse customers kitna owe karte hain
- Customer-wise ledger – ek customer ka poora hisaab
- Staff performance – kisne kitne orders handle kiye
- Monthly P&L summary (revenue only – no expense tracking in v1)
- Overdue invoices list – payment nahi aayi jinki
- Date range filter for all reports

8.3 Expense Module (Optional – v2)

Version 1 mein sirf revenue track hoga. Version 2 mein fabric purchase, salary, rent jaise expenses bhi add honge.

9. Messaging & Notifications

9.1 Customer Messaging

Staff customers ko directly app se message bhej sakta hai. Koi payment gateway nahi – sirf messages ke liye SMS/WhatsApp integration:

Message Type	Trigger / Use
Order Ready Notification	Jab status 'Ready' ho – 'Aapka kapda taiyar hai'
Payment Reminder	Due amount baki hai – gentle reminder
Delivery Confirmation	Order deliver hone ke baad – thank you message
Trial Reminder	Trial date aayi – remind karo customer ko
Custom Message	Staff manually kuch bhi type karke bheje
Advance Payment Receipt	Advance lene ke baad confirmation
Birthday/Anniversary Wish	Optional – date of birth se auto-trigger

9.2 Message Templates

- Admin predefined message templates bana sakta hai
- Templates mein variables use ho sakte hain: {customer_name}, {amount}, {balance}, {order_id}, {shop_name}
- Staff template select karo, variables auto-fill ho, send karo
- Sent messages ki history maintain hoti hai customer profile mein

9.3 Channel Options

- SMS – direct SMS via device (Android SMS intent)
- WhatsApp – WhatsApp deep link se message open hoga (manual send required by staff)
- In-App notification for staff alerts (internal only)

10. UI/UX Design – Responsive Layout

10.1 Device Strategy

Device Type	Layout Strategy
Android Phone (Portrait)	Single column, bottom navigation bar, large touch targets, full-width cards
Android Tablet Landscape	Two-panel layout – list on left (40%), detail on right (60%)
iPad / iOS Tablet	Same as Android tablet layout
iPhone Portrait	Same as Android phone layout

10.2 Navigation Structure

Phone mein bottom navigation bar hogi 5 main tabs ke saath:

- Home / Dashboard
- Customers
- Orders
- Billing / Accounts
- More (Settings, Staff, Reports, Messages)

Tablet mein left side navigation rail hogi jo always visible rahegi landscape mode mein.

10.3 Key UI Components

- Material Design 3 (Material You) – Flutter ka default modern theme
- Custom brand colors – deep blue, teal accents
- Dark mode support
- Large buttons aur input fields for easy touch
- Customer quick-search bar on all relevant screens
- Status chips with color coding for orders
- Summary cards on dashboard with animations
- Pull-to-refresh on all list screens
- Infinite scroll for large customer/order lists

10.4 Print Screen Layout

- PDF preview screen before printing
- Scale and zoom preview
- Choose printer: Bluetooth thermal printer ya share PDF
- Print history log

11. Technical Architecture

11.1 Technology Stack

Component	Technology & Details
Frontend Framework	Flutter 3.x (Dart)
State Management	Provider + Riverpod (recommended)
Local DB (Offline)	SQLite via drift/sqflite package
API Communication	Dio HTTP client with interceptors
Auth	JWT tokens – stored in secure storage
PDF Generation	pdf package (flutter) – custom layouts
Printing	printing + flutter_blue (Bluetooth printer)
SMS/WhatsApp	url_launcher (intent-based)
Image Handling	image_picker + cached_network_image
Notifications	flutter_local_notifications
Backend Framework	Java Spring Boot 3.x
Database	MySQL 8.x (production) / H2 (dev/test)
ORM	JPA / Hibernate
Authentication	Spring Security + JWT (jjwt library)
API Documentation	Swagger / OpenAPI 3.0
Build Tool	Maven / Gradle
Hosting	VPS / AWS / DigitalOcean (backend)
CI/CD	GitHub Actions (optional)

11.2 Database – Core Tables

Table Name	Key Fields	Purpose
users	id, name, phone, email, role_id, pin_hash, is_active	Staff aur admin accounts
roles	id, name, permissions_json	RBAC roles management
customers	id, name, phone, whatsapp, address, type, notes	Customer master data
measurement_categories	id, name, description, is_active	Garment categories
measurement_fields	id, category_id, field_name, field_type, unit, sort_order	Dynamic fields

measurements	id, customer_id, category_id, taken_by, taken_at	Measurement header
measurement_values	id, measurement_id, field_id, value	Actual naap values
orders	id, customer_id, order_no, items_json, total, advance, status, delivery_date	Order management
invoices	id, order_id, invoice_no, subtotal, discount, tax, total	Billing
payments	id, invoice_id, amount, mode, date, received_by, notes	Payment transactions
message_templates	id, name, content, variables	Reusable message templates
messages_sent	id, customer_id, template_id, content, sent_at, sent_by	Message history
shop_settings	id, key, value	Shop configuration

11.3 API Design – Key Endpoints

Method	Endpoint	Auth	Description
POST	/api/auth/login	Public	User login – returns JWT
GET	/api/customers	Staff+	All customers with pagination
POST	/api/customers	Staff+	Create new customer
PUT	/api/customers/{id}	Staff+	Update customer info
GET	/api/customers/{id}/history	Staff+	Full customer history
GET	/api/measurement-categories	Staff+	Get all garment types
POST	/api/measurement-categories	Admin/Mgr	Add new category
POST	/api/measurement-fields	Admin/Mgr	Add field to category
GET	/api/customers/{id}/measurements	Staff+	Customer's all measurements
POST	/api/measurements	Staff+	Save new measurement
GET	/api/orders	Staff+	All orders with filters
POST	/api/orders	Staff+	Create new order
PATCH	/api/orders/{id}/status	Staff+	Update order status
POST	/api/invoices	Staff+	Generate invoice
POST	/api/payments	Staff+	Record payment
GET	/api/reports/dues	Admin/Mgr	Pending dues report
GET	/api/reports/collection	Admin/Mgr	Daily collection report
GET	/api/users	Admin	Staff management
POST	/api/messages/send	Staff+	Send SMS/WhatsApp message

12. Security & Data Protection

12.1 Authentication & Authorization

- JWT token based authentication – access token (1 hour) + refresh token (30 days)
- Role permissions check both frontend (UI hide) aur backend (API reject) pe
- PIN-based quick login for staff on shared devices
- Auto logout after configurable idle time
- Failed login attempts limit (5 attempts → 30 min lockout)

12.2 Data Security

- All API calls HTTPS only (SSL/TLS)
- Passwords bcrypt hash se store honge
- Sensitive data (payment amounts) encrypted at rest
- Database backups – daily automated backup
- Audit log – koi bhi delete ya payment change kare to log maintain ho

13. Settings & Configuration

13.1 Shop Settings

Setting	Description
Shop Name	Bill aur print pe aayega
Logo	Shop ka logo upload
Address & Contact	Print pe aane ke liye
GST Number	Optional – invoice pe
Currency Symbol	Default: ₹
Default Unit	CM ya Inch
Tax Rate	GST % – 0 rakhne se disabled
Invoice Prefix	INV, BILL, etc.
Order No. Prefix	ORD, etc.
Business Hours	Optional – reminder ke liye
WhatsApp Number	Shop ka WhatsApp

13.2 Staff Management

- Admin staff add, edit, deactivate kar sakta hai
- Role assign karna
- Password reset karna
- Login history dekhna
- Staff ka work report dekhna – orders, collections, etc.

14. Development Phases & Timeline

Phase	Duration	Deliverables	Priority
1	Week 1-2	Project setup, DB schema, Auth module, Basic API skeleton	Critical
2	Week 3-4	Customer module (CRUD + search), Measurement module with dynamic fields	Critical
3	Week 5-6	Orders module, Status workflow, Dashboard widgets	Critical
4	Week 7-8	Billing, Invoice, Partial payment tracking	Critical
5	Week 9-10	Accounts reports, Print (PDF + Bluetooth), Messaging	High
6	Week 11-12	Staff management, Role permissions, Settings	High
7	Week 13-14	UI polish, Tablet layout, Dark mode, Performance	Medium
8	Week 15-16	Testing, Bug fixes, UAT, App store build	Critical

15. Future Scope (Version 2.0)

15.1 Features for Next Version

Feature	Description
Expense Tracking	Fabric, accessories, salary expenses record karein
Inventory Management	Fabric stock maintain karo
Barcode/QR System	Order ke liye QR code – customer scan kare status dekhne ke liye
WhatsApp Business API	Automated message sending without manual click
Payment Gateway	UPI/Razorpay integration for online payments
Multi-Branch Support	Ek account se multiple shop locations
Customer App (Lite)	Customer khud apna naap aur order status dekh sake
Analytics Dashboard	Advanced charts – busiest month, top customers, etc.
Cloud Sync + Offline Mode	Full offline kaam kare, sync jab internet aaye
AI Suggestions	Customer ke based pe suggest karo – 'Iska pichhla order 2 month mein tha'

16. Glossary

Term	Meaning
Naap	Measurement – customer ka size/dimension
RBAC	Role-Based Access Control – role se permissions decide hote hain
JWT	JSON Web Token – secure authentication ke liye
ORM	Object Relational Mapping – Java objects ko DB tables se map karna
API	Application Programming Interface – frontend-backend communication
PDF	Portable Document Format – print ke liye
UPI	Unified Payments Interface – Indian digital payment
GST	Goods and Services Tax – India ka tax system
UAT	User Acceptance Testing – actual users se testing
CRUD	Create, Read, Update, Delete – basic database operations

Document End – SilaiPro PRD v1.0

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